Construct a complete truth table for each of the following. Include intermediate columns.

1. $\sim(p \to q)$

2. $\sim(p \vee q) \vee \sim p$

3. $\sim(\sim p \vee \sim q)$

4. $p \vee (q \wedge \sim r)$

5. $p \vee (q \to r)$

6. $(q \vee r) \leftrightarrow (r \wedge q)$